

Definition 19.1

Let R be a ring. An *(both sided) ideal* of R is a subring $I \subseteq R$ such that for any $a \in I$ and $r \in R$ we have $ra \in I$ and $ar \in I$.

We write $I \triangleleft R$ to denote that I is an ideal of R .

Theorem 19.2

Let R be a rings and $I \triangleleft R$. Let $a_1, a_2, b_1, b_2 \in R$ be elements such that $a_1 + I = a_2 + I$ and $b_1 + I = b_2 + I$. Then $(a_1 b_1) + I = (a_2 b_2) + I$.

Definition 19.3

Let R be a ring and let $I \triangleleft R$. The *quotient ring* R/I is the ring defined as follows:

- Elements of R/I are cosets $a + I$ for $a \in R$.
- Addition: $(a + I) + (b + I) = (a + b) + I$.
- Multiplication: $(a + I) \cdot (b + I) = (ab) + I$.

Definition 19.4

Let R be a commutative ring with unity. A *prime ideal* of R is an ideal $I \triangleleft R$ such that $I \neq R$ and if $ab \in I$ then either $a \in I$ or $b \in I$.

Theorem 19.5

If R is a commutative ring with unity and $I \triangleleft R$ then I is a prime ideal if and only if R/I is an integral domain.

Definition 19.6

Let R be a commutative ring with unity. A *maximal ideal* of R is an ideal $I \triangleleft R$ such that $I \neq R$ if $J \triangleleft R$ is any other ideal such that $J \neq R$ and $I \subseteq J$, then $I = J$.

Theorem 19.7

If R is a commutative ring with unity and $I \triangleleft R$ then I is a maximal ideal if and only if R/I is a field.

Theorem 19.8

If R is a commutative ring with unity then every maximal ideal of R is a prime ideal.